Assignments

Q1. Explain with sketches the different types of Cams and followers?

Q2. Draw the displacement, velocity and acceleration diagrams for a follower with following motion;

(a) Simple Harmonic motion (SHM)
(b) Uniform acceleration and retardation.

Q3. ~~A~~ Single block brake is shown in fig. The diameter of the drum is 250 mm and the angle of contact is 90°. If the operating force of 700 N is applied at the end of a lever and the coefficient of friction between the drum and the lining is 0.35, determine the torque that may be transmitted by the block brake.

Q4. A single block brake ~~as shown in figure~~ has the drum diameter 250 mm. The angle of contact is 90° and the coefficient of friction between the drum and the lining is 0.35. If the torque transmitted by the brake is 80 Nm, find the force required to operate the brake.

Q5. A single block brake ~~is shown in figure~~. The dia. of the drum is 400 mm and the angle of contact is 90°. If the operating force of 6 kN is applied at the end of a lever and the coefficient of friction between the drum and the lining is 0.30, determine the torque that may be transmitted by the block brake.

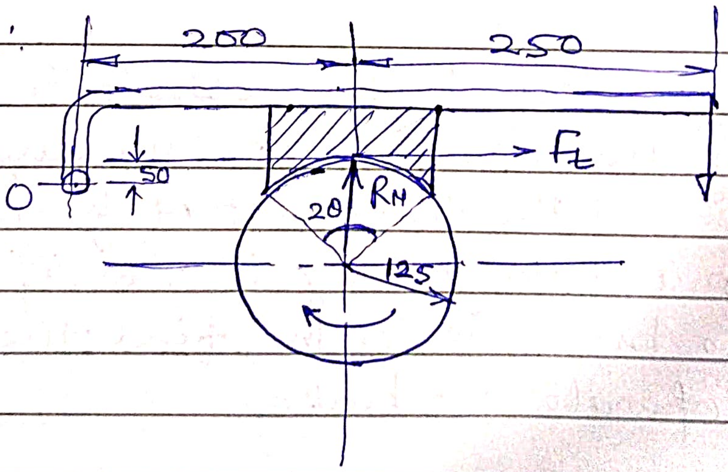
Q 6. A Cam is to be designed for a knife edge follower with the following data:

1. Cam lift = 40 mm during 90° of Cam rotation with SHM.
2. Dwell for the next 30°
3. During next 60° of Cam rotation, the follower returns to its original position with SHM.
4. Dwell during the remaining 180°

(a) Draw the profile of Cam when line of stroke of follower passes through the axis of the Cam shaft.
The radius of the base circle of the Cam is 40 mm.

(b) The line of stroke is offset 20 mm from axis of Cam shaft.

Figure for Q3.



figure

Figure for Q5.

Q6. Explain single block or shoe brake when the line of action of tangential force passes through a distance "a" above the fulcrum O, and the wheel rotates clockwise and Counter clockwise.

Q7. Explain materials for Brake lining.